

Chemical Bonding
TPA-1

Inorganic Chemistry

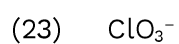
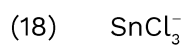
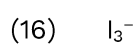
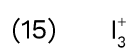
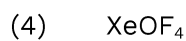
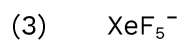
Topic Practice Assignment

Prepared by: Rajeev Sir

Q1. Draw Lewis dot structures of following species.

- | | |
|----------------------------------|------------------------------------|
| 1. CO | 16. HNC |
| 2. CO ₂ | 17. SiF ₄ |
| 3. NO ₂ ⁻ | 18. SnCl ₂ |
| 4. NO ₃ ⁻ | 19. BF ₄ ⁻ |
| 5. CCl ₃ ⁻ | 20. BH ₄ ⁻ |
| 6. COCl ₂ | 21. BeF ₄ ²⁻ |
| 7. N ₃ ⁻ | 22. H ₃ O ⁺ |
| 8. O ₃ | 23. SO ₃ |
| 9. CH ₃ Cl | 24. SO ₂ |
| 10. NH ₄ ⁺ | 25. CO ₃ ²⁻ |
| 11. NH ₂ Cl | 26. NO ₂ Cl |
| 12. OCN ⁻ | 27. NOCl |
| 13. CN ⁻ | 28. F ₂ O |
| 14. SCN ⁻ | 29. SO ₄ ²⁻ |
| 15. HCN | 30. PO ₄ ³⁻ |
| | 31. H ₂ O ₂ |
| | 32. O ₂ F ₂ |
| | 33. S ₂ Cl ₂ |

Q2. Draw the structure of the following molecules / ions.



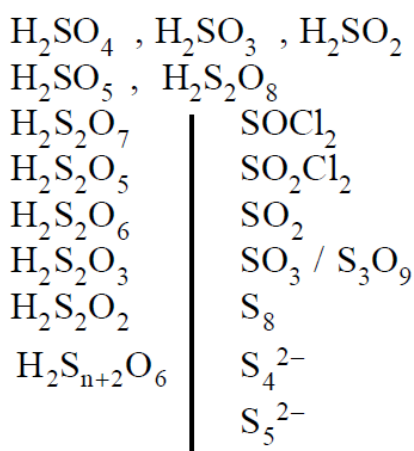
Q3- Draw structures of following compounds.

Structure of V group.

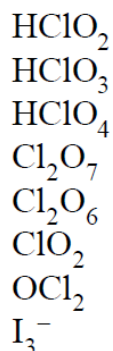
N_2O , NO , N_2O_4 , N_2O_5 , N_2O_3 , HNO_3 , HNO_2 , HNO_4 , $(\text{HNO})_2$, N_5^+ , N_3H , NH_3 , N_2H_4
Symm & unsymm

P_4O_6 , P_4O_{10} , P_4S_6 , P_4S_{10} , H_3PO_4 , H_3PO_3 , H_3PO_2 , $\text{H}_4\text{P}_2\text{O}_7$, H_3PO_5 , $\text{H}_4\text{P}_2\text{O}_8$, $\text{H}_4\text{P}_2\text{O}_6$
Diphosphoric acid , Triphosphoric acid, White P, Red P, Black P.

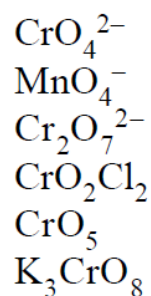
VI group



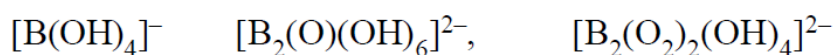
VII group



d-Block



group-13



Inorganic benzene, Inorganic graphite.

Boric acid, Borax .